

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru-3bb728174 | github.com/zac-09 | Kampala, Uganda

PROFESSIONAL SUMMARY

Senior Backend Engineer with 4+ years building production Node.js and TypeScript systems at scale. Led zero-downtime migration of 20+ applications from Parse/MongoDB to Firebase/GCP, designing real-time data sync pipelines using Pub/Sub that process millions of records. Expert in AWS and cloud infrastructure with proven track record of architecting scalable backend services, microservices with Kubernetes, and serverless solutions. Experienced working directly with CTOs on architecture decisions and building AI-ready data platforms. Passionate about joining early-stage teams to build robust, scalable infrastructure for AI-powered products.

CORE COMPETENCIES

Node.js & TypeScript

Backend Services Design

AWS & Cloud Infrastructure

Real-time Data Pipelines

Kubernetes & Serverless

Production Systems

Firebase & GCP

Scalable Architecture

WORK EXPERIENCE

MTailor

September 2022 – Present

Software Engineer · US Remote

- **Architected real-time two-way sync** between MongoDB and Firestore using Node.js and Google Pub/Sub for message processing — enabling seamless data migration without service interruption
- **Led end-to-end migration** of 20+ production applications from Parse/MongoDB to Firebase/GCP, ensuring zero downtime while reporting directly to the CTO
- **Built scalable backend services** including a 3D visualization feature using video overlay and ffmpeg, increasing buyer conversion
- **Optimized cloud infrastructure costs** by migrating all services from AWS to GCP, saving \$5,000/month
- Implemented Express Shipping feature generating additional \$40 revenue per order

MTailor

March 2022 – July 2022

Contractor · US Remote

- **Designed and built migration scripts** in Node.js to read data from MongoDB and write to Firestore with complex processing logic for production data transformation
- Authored comprehensive documentation for engineers on Firebase SDK integration (Node.js and Python)
- Trained operations team on Firebase Dashboard and backup procedures

Dr Wealth

August 2021 – January 2022

Software Engineer · Singapore Remote

- **Built scalable data processing jobs** querying over 2 million Firestore records to maintain real-time stock price data from Morningstar APIs
- Extended backend APIs using Firebase Cloud Functions and Express.js for serverless architecture
- Developed responsive UI components for PWA using React and Tailwind

Mind2matter.co

January 2022 – July 2022

Software Engineer · US Remote

- **Built backend systems for DeFi applications** using Web3 and Node.js, reporting directly to CTO
- Delivered production React UIs from Figma designs under tight deadlines

CodeBits

January 2020 – July 2021

Software Engineer · Uganda Remote

- **Architected microservices backend** for FIDA Uganda's case management system using Apache Kafka, Docker, and Kubernetes
- **Led team of 4 developers** building legal tech systems for NGOs across Uganda
- Built cross-platform mobile app using React Native and Expo with Firebase push notifications
- Developed USSD service communicating with Node.js backend via gRPC for legal aid providers

PROJECTS

Real-time Data Sync Pipeline Production

Built live MongoDB to Firestore bidirectional sync pipeline supporting parallel production traffic during zero-downtime migration. Handles message ordering, conflict resolution, and data transformation at scale.

Node.js · Google Pub/Sub · Firestore · MongoDB

Healthcare WhatsApp Chatbot AI Integration

Integrated OpenAI and Google Vision APIs for prescription OCR via WhatsApp Business API. Demonstrates AI/LLM integration in production backend systems.

Node.js · OpenAI API · Google Vision · WhatsApp Business API

Microservices Case Management Platform Kubernetes

Designed and deployed containerized microservices architecture for legal aid case management serving multiple NGOs on shared infrastructure.

Kubernetes · Docker · Apache Kafka · Node.js

EDUCATION

Bachelor of Science in Computer Engineering — [Makerere University](#) 2018 – 2022

Kampala, Uganda

TECHNICAL SKILLS

Backend: Node.js, TypeScript, Express.js, Python **Cloud:** AWS (S3, EC2, EBS), GCP, Firebase, Serverless

Infrastructure: Kubernetes, Docker, Pub/Sub, Kafka **Databases:** MongoDB, Firestore, SQL **Frontend:** React, React Native, Tailwind CSS

Tools: Git, Linux, gRPC